

A trend analysis of breast cancer incidence rates in the United States from 2000 to 2009 shows a recent increase

Ningqi Hou · Dezheng Huo

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Abstract Recent reports have shown that the breast cancer incidence rate in the US stabilized after a sharp reduction in 2002 and 2003. It is important to continue monitoring breast cancer incidence rates according to age group, race/ethnicity, estrogen receptor (ER) status, and tumor stage. Age-standardized breast cancer incidence rates were calculated using data from the surveillance, epidemiology, and end results 18 registries from 2000 to 2009, for 677,774 female breast cancer patients aged 20 and above. Jointpoint regression models were used to fit a series of joined straight lines on a log scale to annual age-standardized rates. The incidence rates of all breast cancer significantly increased for non-Hispanic blacks from 2005 to 2009 (annual percentage change, APC = 2.0 %, $p = 0.01$) and Asian/Pacific Islanders from 2000 to 2009 (APC = 1.2 %, $p = 0.02$). Since 2004, incidence rates in women aged 40–49 years significantly increased for most racial/ethnic groups (overall APC = 1.1 %, $p = 0.001$). The incidence rate of carcinoma in situ significantly increased in all racial/ethnic groups, with an APC range from 2.3 to 3.0 % ($p < 0.005$). The localized breast cancer incidence significantly increased in non-Hispanic blacks (APC = 1.3 %, $p = 0.004$) and Asians (APC = 1.2 %, $p = 0.03$). ER-positive breast cancer significantly increased in almost all age/race sub-groups after 2005 (APC by race: non-Hispanic whites 1.5 %, non-Hispanic blacks 4.3 %, Asian/Pacific Islanders 1.7 %, and Hispanics 1.8 %; all p values < 0.05), while ER-negative breast cancer decreased in most sub-groups (APC by race:

non-Hispanic whites—3.9 %, non-Hispanic blacks—3.7 %, Asian/Pacific Islanders—1.5 %, and Hispanics—4.3 %; all p values < 0.05). Recently the incidence of breast cancer appears to be increasing in certain subgroups, including ER-positive, early-stage breast cancers, in particular among non-Hispanic blacks and Asian/Pacific Islanders. Further studies are warranted to examine possible reasons for these changes, such as changes in mammography screening methods and risk factors prevalence.

Keywords Breast cancer · Incidence · SEER · Race · Hormone receptors · Subgroup

Background

There was a sharp decrease in breast cancer incidence from 2002 to 2003 in the United States, evident in white women ≥ 50 years of age and for estrogen receptor (ER) positive tumors. This has been attributed to the reduced use of hormone replacement therapy (HRT) after the July 2002 publication from the Women's Health Initiative [1–4]. Following the sharp decrease, it was reported that a decreasing trend continued into 2005 [5, 6]. The most recent publications observed that the decline in cancer incidence ceased [7], and the rate stabilized from 2004 to 2008 [8], with ER-positive breast cancer being projected to increase [9]. In this brief report, we examined the breast cancer incidence rates from 2000 to 2009 in the United States, using the updated data from 18 registries of the Surveillance, Epidemiology, and End Results (SEER) program. We present the rates by age group, race/ethnicity, estrogen receptor status, and stage of tumor, in order to inform the underlying reasons for the changes.

N. Hou · D. Huo (✉)
Department of Health Studies, University of Chicago, 5841
South Maryland Avenue, MC 2007, Chicago, IL 60637, USA
e-mail: dhuo@health.bsd.uchicago.edu

Methods

Using data from the SEER 18 registries [10], we examined the overall and subgroup trends of breast cancer incidence in the United States from 2000 to 2009. We obtained patient and population data from SEER 9 Registries

Database, covering Atlanta, Connecticut, Detroit, Hawaii, Iowa, New Mexico, San Francisco-Oakland, Seattle-Puget Sound, and Utah; the SEER 4 Registries Database that includes Los Angeles, San-Jose Monterey, Rural Georgia, and Alaska Native Tumor Registry; and the SEER 5 Registries Database covering Greater California, Kentucky,

Table 1 Characteristics of breast cancer patients in the United States: data from the Surveillance, Epidemiology, and End Results 18 registries database for cancers diagnosed between 2000 and 2009

N	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	Total
Race/Ethnicity											
White, non-Hispanic	50,672	51,808	50,962	48,866	49,272	47,979	49,553	51,196	52,204	52,985	505,497
	77.8 %	77.6 %	75.6 %	74.8 %	74.1 %	73.5 %	72.8 %	72.1 %	71.8 %	71.8 %	74.6 %
Black, non-Hispanic	5,917	5,970	6,151	6,257	6,544	6,090	6,886	7,343	7,426	7,831	66,415
	9.1 %	8.9 %	9.2 %	9.7 %	9.9 %	9.4 %	10.2 %	10.4 %	10.3 %	10.6 %	9.8 %
Asian/Pacific Islander	3,497	3,859	4,153	3,927	4,281	4,435	4,617	5,001	5,474	5,477	44,721
	5.4 %	5.8 %	6.2 %	6.1 %	6.5 %	6.9 %	6.9 %	7.1 %	7.6 %	7.4 %	6.6 %
Hispanic	4,755	4,877	5,146	5,283	5,529	5,992	6,101	6,527	6,969	7,140	58,319
	7.3 %	7.3 %	7.7 %	8.2 %	8.4 %	9.3 %	9.1 %	9.3 %	9.6 %	9.7 %	8.6 %
Native American	241	247	253	274	287	277	290	310	318	325	2,822
	0.4 %	0.4 %	0.4 %	0.4 %	0.4 %	0.4 %	0.4 %	0.4 %	0.4 %	0.4 %	0.4 %
Age of diagnosis											
20–39	3,382	3,482	3,170	3,222	3,338	3,110	3,196	3,134	3,335	3,204	32,573
	5.2 %	5.2 %	4.8 %	5.0 %	5.1 %	4.8 %	4.7 %	4.5 %	4.6 %	4.3 %	4.8 %
40–49	11,775	11,840	11,934	12,211	12,593	12,182	12,844	12,997	12,838	12,952	124,166
	18.1 %	17.7 %	17.9 %	18.9 %	19.1 %	18.8 %	19.0 %	18.5 %	17.7 %	17.6 %	18.3 %
50–69	29,328	30,526	31,121	29,875	30,585	30,538	31,844	34,269	35,881	36,906	320,873
	45.1 %	45.7 %	46.7 %	46.2 %	46.4 %	47.2 %	47.2 %	48.7 %	49.6 %	50.0 %	47.3 %
70+	20,596	20,907	20,435	19,292	19,392	18,938	19,554	19,971	20,331	20,690	200,106
	31.7 %	31.3 %	30.7 %	29.9 %	29.4 %	29.2 %	29.0 %	28.4 %	28.1 %	28.1 %	29.5 %
Unknown	1	6	5	7	5	5	9	6	6	6	56
	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %
Stage of tumor											
In Situ	11,322	11,963	12,187	12,166	12,583	12,396	13,187	13,940	14,865	15,139	129,748
	17.4 %	17.9 %	18.3 %	18.8 %	19.1 %	19.1 %	19.6 %	19.8 %	20.5 %	20.5 %	19.1 %
Localized	32,501	33,209	33,103	31,533	32,515	32,064	33,360	34,800	35,668	36,930	335,683
	49.9 %	49.7 %	49.7 %	48.8 %	49.3 %	49.5 %	49.5 %	49.5 %	49.3 %	50.1 %	49.5 %
Regional	16,398	16,885	16,752	16,363	16,216	15,894	16,319	16,860	17,135	16,902	165,724
	25.2 %	25.3 %	25.1 %	25.3 %	24.6 %	24.5 %	24.2 %	24.0 %	23.7 %	22.9 %	24.5 %
Distant	3,142	3,154	3,162	3,097	3,346	3,278	3,414	3,587	3,672	3,702	33,554
	4.8 %	4.7 %	4.7 %	4.8 %	5.1 %	5.1 %	5.1 %	5.1 %	5.1 %	5.0 %	5.0 %
Unknown	1,719	1,550	1,461	1,448	1,253	1,141	1,167	1,190	1,051	1,085	13,065
	2.6 %	2.3 %	2.2 %	2.2 %	1.9 %	1.8 %	1.7 %	1.7 %	1.5 %	1.5 %	1.9 %
ER status											
ER+	33,682	35,360	35,905	36,274	41,052	42,668	45,882	49,223	51,924	54,389	426,359
	51.8 %	53.0 %	53.9 %	56.2 %	62.3 %	65.9 %	68.0 %	69.9 %	71.7 %	73.7 %	62.9 %
ER–	9,336	9,562	9,737	10,182	11,950	12,013	12,395	12,480	12,679	12,023	112,357
	14.3 %	14.3 %	14.6 %	15.8 %	18.1 %	18.6 %	18.4 %	17.7 %	17.5 %	16.3 %	16.6 %
Unknown	22,064	21,839	21,023	18,151	12,911	10,092	9,170	8,674	7,788	7,346	139,058
	33.9 %	32.7 %	31.5 %	28.1 %	19.6 %	15.6 %	13.6 %	12.3 %	10.8 %	10.0 %	20.5 %

New Jersey, Greater Georgia, and Louisiana. Data of the 3rd and 4th quarters of 2005 in the Louisiana registry were excluded because of Hurricane Katrina.

Statistical analysis

We restricted the analysis to women aged 20 years or above, and categorized age groups as 20–39, 40–49, 50–69, and 70+ years. We categorized race/ethnicity subgroups as non-Hispanic white, non-Hispanic black, Asian/Pacific Islander, Hispanic, and others (data not shown due to small sample size).

Estrogen receptor status was missing for nearly 20 % of the records. Since hormone receptors are important biomarkers for breast carcinoma and a large portion of missing data can be a major source of bias in rate calculation, we employed the multiple imputation (MI) method to account for the missing values of ER status. Using the chained equation methods [11, 12], we generated 15 complete datasets with ER, progesterone receptor status, age, race/ethnicity, histological type, grade, SEER registry, year of diagnosis, tumor size, number of positive lymph nodes, SEER historic stage, vital status, and survival time in the imputation models.

Age-adjusted incidence rates were calculated based on the 2000 US standard population. We used the Joinpoint Regression Program 3.5.4 to fit a series of joined straight lines on a log scale to the trends of annual, age-adjusted rates, with the fitted slopes indicating annual percentage change (APC) [13]. All statistical analyses were performed using STATA 12.0, with the exception of the use of SAS 9.3 to read in the original SEER datasets.

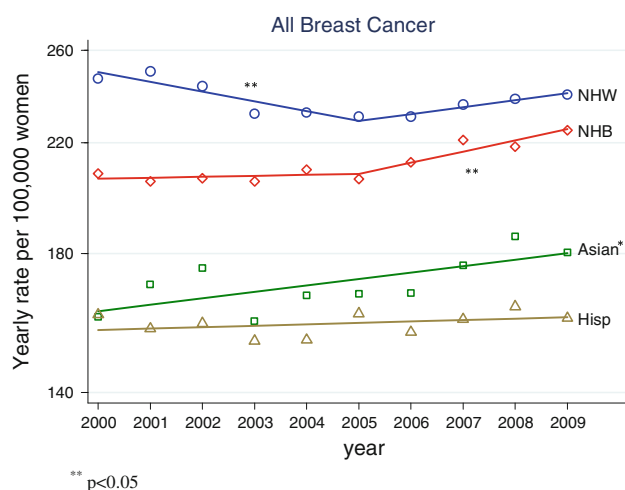
Results

The combined SEER registries from 2000 through 2009 included 677,774 female breast cancer patients 20 years or older (Table 1). The proportion of missing ER status has decreased from 33.9 % in 2000 to 10.0 % in 2009. Figure 1 shows the trend of incidence rates of all breast cancer in the United States between 2000 and 2009, stratified by race/ethnicity. The rate for non-Hispanic whites decreased significantly from 246.9 per 100,000 in 2000 to 228.8 per 100,000 in 2005 (APC = -1.8% , $p = 0.01$), but then rose to 240.6 per 100,000 in 2009 (APC = 1.3% , $p = 0.11$). The incidence rate for non-Hispanic blacks was stable between 2000 and 2005 (APC = 0.2% , $p = 0.7$), but significantly increased from 206.0 per 100,000 in 2005 to 224.9 per 100,000 in 2009 (APC = 2.0% , $p = 0.01$). Throughout the decade, the incidence rate increased significantly for Asian/Pacific Islanders (APC = 1.2% , $p = 0.02$), but remained stable for Hispanic women (APC = 0.41% , $p = 0.3$).

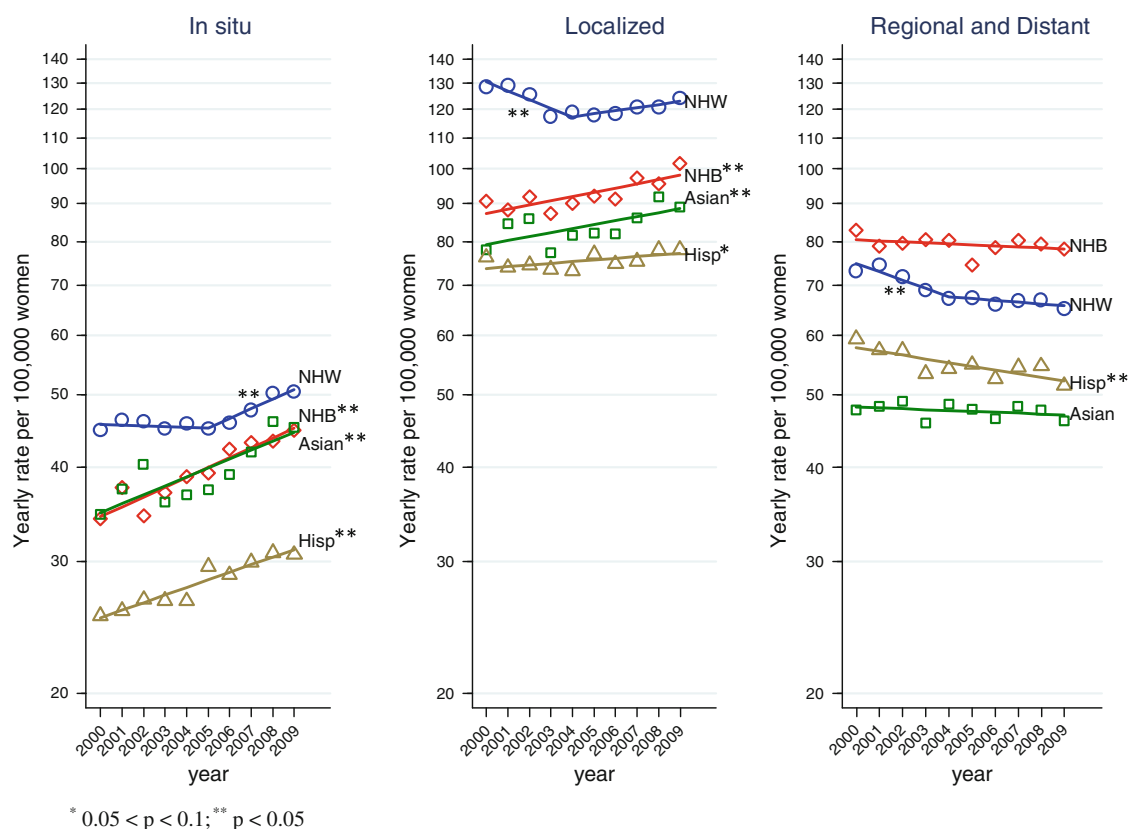
Taking tumor stage into consideration, Fig. 2 shows that the incidence of in situ breast cancer significantly increased in all racial/ethnic groups while for non-Hispanic whites, the rate only increased after 2005 (APC ranges from 2.3 to 3.0 % by racial/ethnic group, p values <0.01). Localized breast cancer significantly increased in non-Hispanic blacks (APC = 1.3% , $p = 0.004$) and Asian/Pacific Islanders (APC = 1.2% , $p = 0.03$). Regional and distant cancers significantly decreased in non-Hispanic whites from 2000 to 2004 (APC = -2.5% , $p = 0.02$) and in Hispanics from 2000 to 2009 (APC = -1.1% , $p = 0.006$).

Stratified by age and race/ethnicity (Fig. 3), the incidence rate increased significantly in all age groups for non-

Fig. 1 Annual incidence rate of all female breast cancer in the United States with Joinpoint Regression (line of trends), by race/ethnicity. Data from the U.S. National Cancer Institute's Surveillance, Epidemiology, and End Results 18 Registries Database for cancers diagnosed between 2000 and 2009



NHW (non-Hispanic white): slope I: -1.8% , $p = 0.01$; slope II: 1.3% , $p = 0.11$;
 NHB (non-Hispanic black): slope I: 0.2% , $p = 0.7$; slope II: 2.0% , $p = 0.01$;
 Asian (Asian/Pacific Islander): 1.2% , $p = 0.02$; Hisp (Hispanics): 0.3% , $p = 0.3$



NHW, non-Hispanic white; NHB, non-Hispanic black; Asian, Asian/Pacific Islander; Hisp, Hispanics

In situ: NHW slope I -0.2 % p = 0.7; slope II 2.9 % p = 0.007; NHB 3.0 % p = 0.00002; Asian 2.7 % p = 0.003; Hispanic 2.3 % p = 0.0002; **Localized:** NHW slope I -2.7 % p = 0.02; slope II 1.0 % p = 0.11; NHB 1.3 % p = 0.004; Asian 1.2 % p = 0.03; Hisp 0.5 % p = 0.051; **Regional & distant:** NHW slope I -2.5 % p = 0.02; slope II -0.5 % p = 0.3; NHB -0.3 % p = 0.3; Asian -0.3 % p = 0.3; Hisp -1.1 % p = 0.006

Fig. 2 Annual incidence rate of female breast cancer in the United States with Joinpoint Regression (line of trends), by stage of tumor and race/ethnicity. Data from the U.S. National Cancer Institute's

Surveillance, Epidemiology, and End Results 18 Registries Database for cancers diagnosed between 2000 and 2009

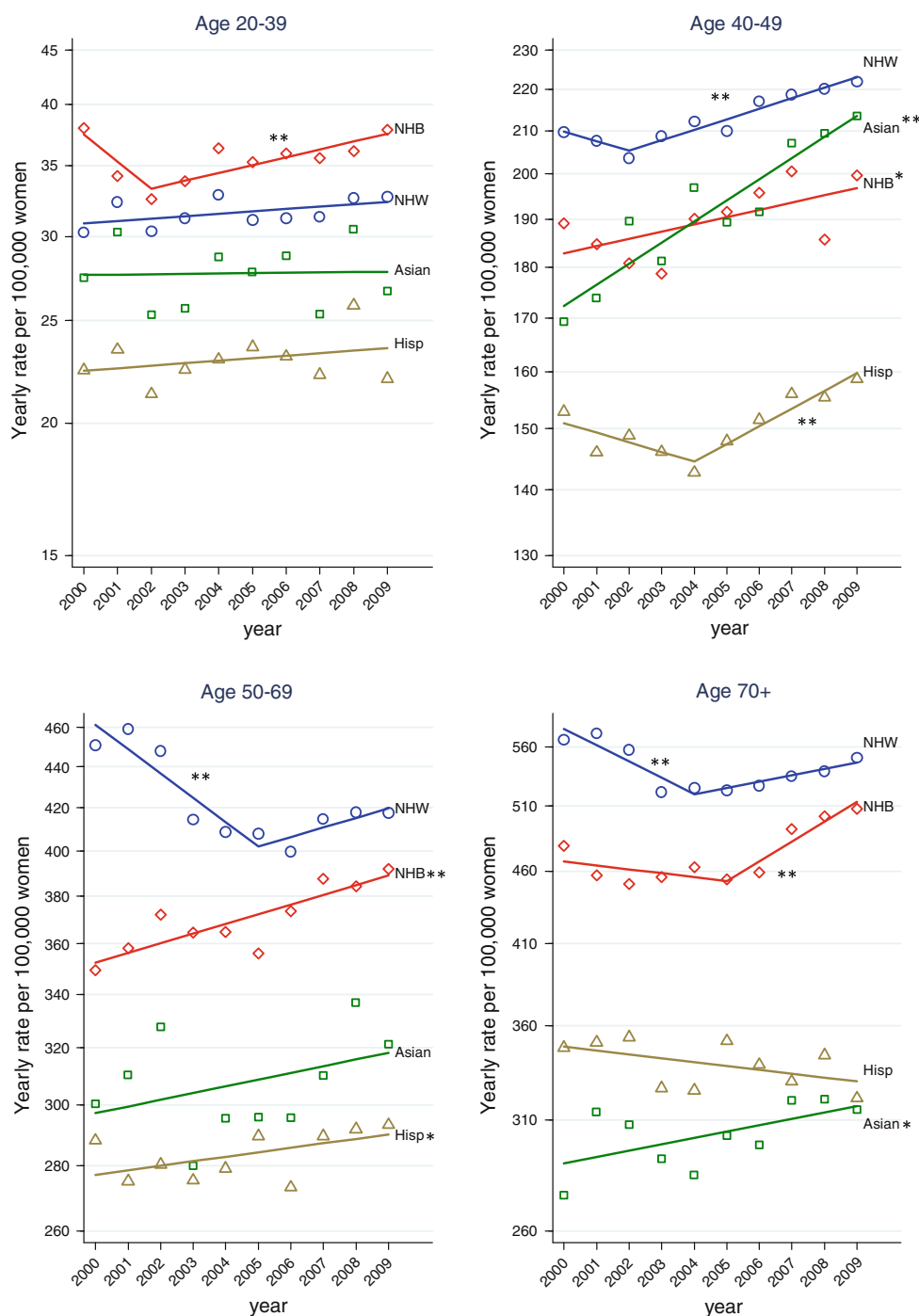
Hispanic blacks, 2005–2009. For Asian/Pacific Islanders, the significant increase only occurred in the age group 40–49 years (APC = 2.3 %, $p < 0.0001$). For this age group (40–49 years), the incidence rates also increased in non-Hispanic whites and Hispanics since 2004 (p values <0.005). For non-Hispanic whites, the incidence decreased significantly for both the age 50–69 group and the age 70+ group ($p < 0.02$) between 2000 and 2004.

Figure 4 presents the temporal trends further stratified by ER status. The incidence of ER-positive breast cancer significantly increased in almost all age and racial/ethnic subgroups after 2005, with annual percent changes particularly high for women of older ages: for non-Hispanic whites, APC = 1.8, 1.5, 1.4, and 1.6 % in age groups 20–39, 40–49, 50–69, and 70+, respectively (p values <0.02); for non-Hispanic blacks, APC = 2.8, 2.4, 2.2, and 4.5 % in age groups 20–39, 40–49, 50–69, and 70+, respectively (p values <0.02). A similar but weaker trend

was observed in Asians and Hispanics, while there was no significant change in incidence in Asians aged 20–39 and Hispanics aged above 70. On the other hand, the incidence of ER-negative breast cancer significantly decreased in most age and racial/ethnic subgroups: all non-Hispanic whites, all non-Hispanic blacks (marginally significant in age group ≥ 70 years old), Asian 20–39 years old, and the Hispanics aged ≥ 40 years old (Fig. 4, second part).

Discussions

Using the SEER 18 registries data, we observed that the incidence of all breast cancer significantly increased in non-Hispanic black and Asian women in recent years (2005–2009). This increasing trend is not significant in non-Hispanic white women, among whom the dramatic decrease in incidence following the 2002 WHI publication



NHW, non-Hispanic white; NHB, non-Hispanic black; Asian, Asian/Pacific Islander; Hisp, Hispanics

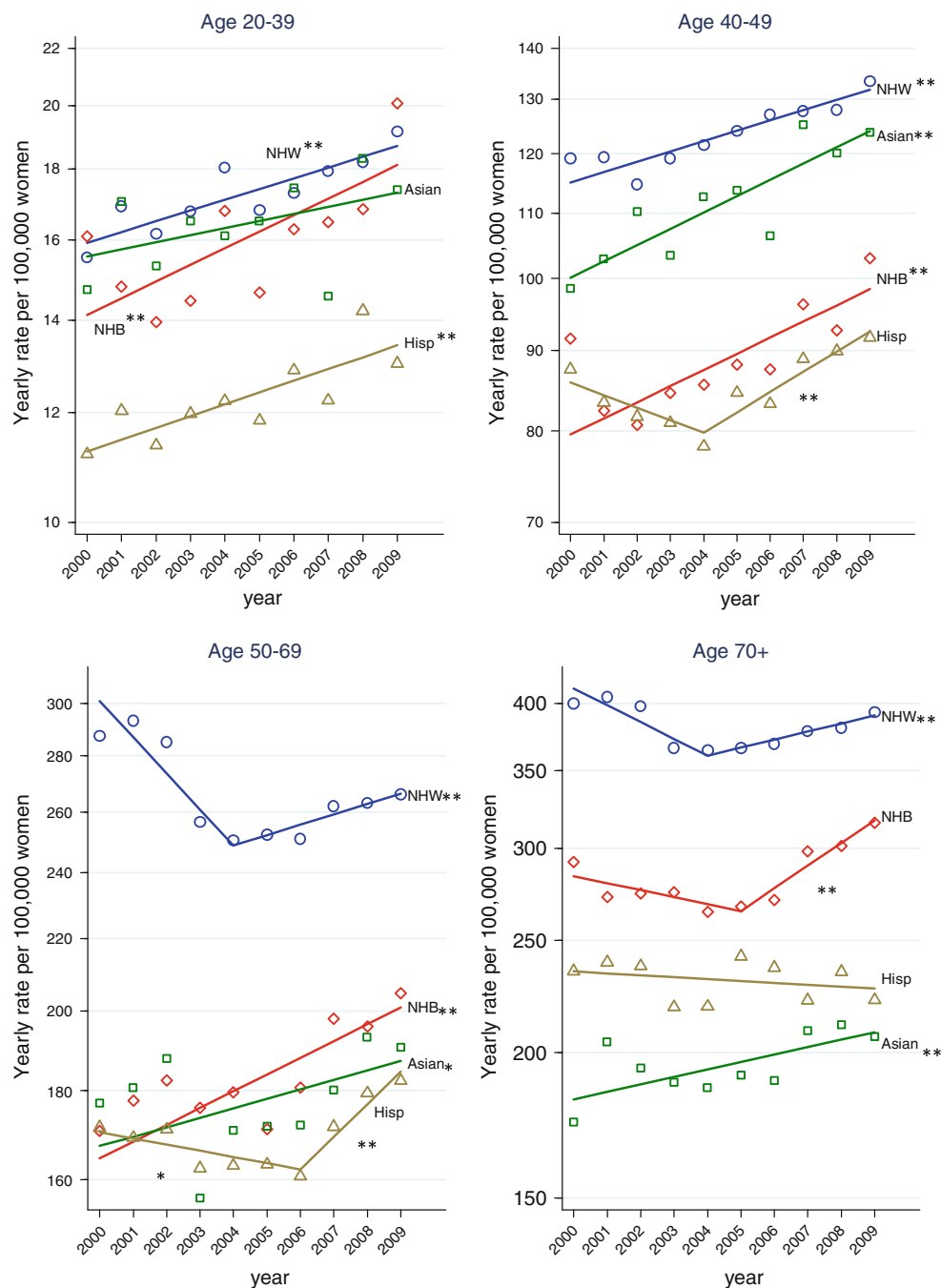
Slopes with $p < 0.1$ (* $0.05 < p < 0.1$; ** $p < 0.05$):

Age 20-39: NHB slope II 1.7 % $p = 0.01$; **Age 40-49:** NHW slope II 1.2 %, $p = 0.0006$; NHB 0.8 % $p = 0.051$; Asian 2.3 % $p < 0.0001$; Hisp slope II 2.0 % $p = 0.004$; **Age 50-69:** NHW slope I -2.8 % $p = 0.01$; NHB 1.1 % $p = 0.001$; Hisp 0.5 % $p = 0.09$; **Age 70+:** NHW slope I -2.6 % $p = 0.02$; NHB slope II 3.1 % $p = 0.02$; Asian 1.0 % $p = 0.06$.

Fig. 3 Annual incidence rate of all female breast cancer in the United States with Joinpoint Regression (line of trends), by age group and race/ethnicity. Data from the U.S. National Cancer Institute's

Surveillance, Epidemiology, and End Results 18 Registries Database for cancers diagnosed between 2000 and 2009

ER+ Breast Cancer



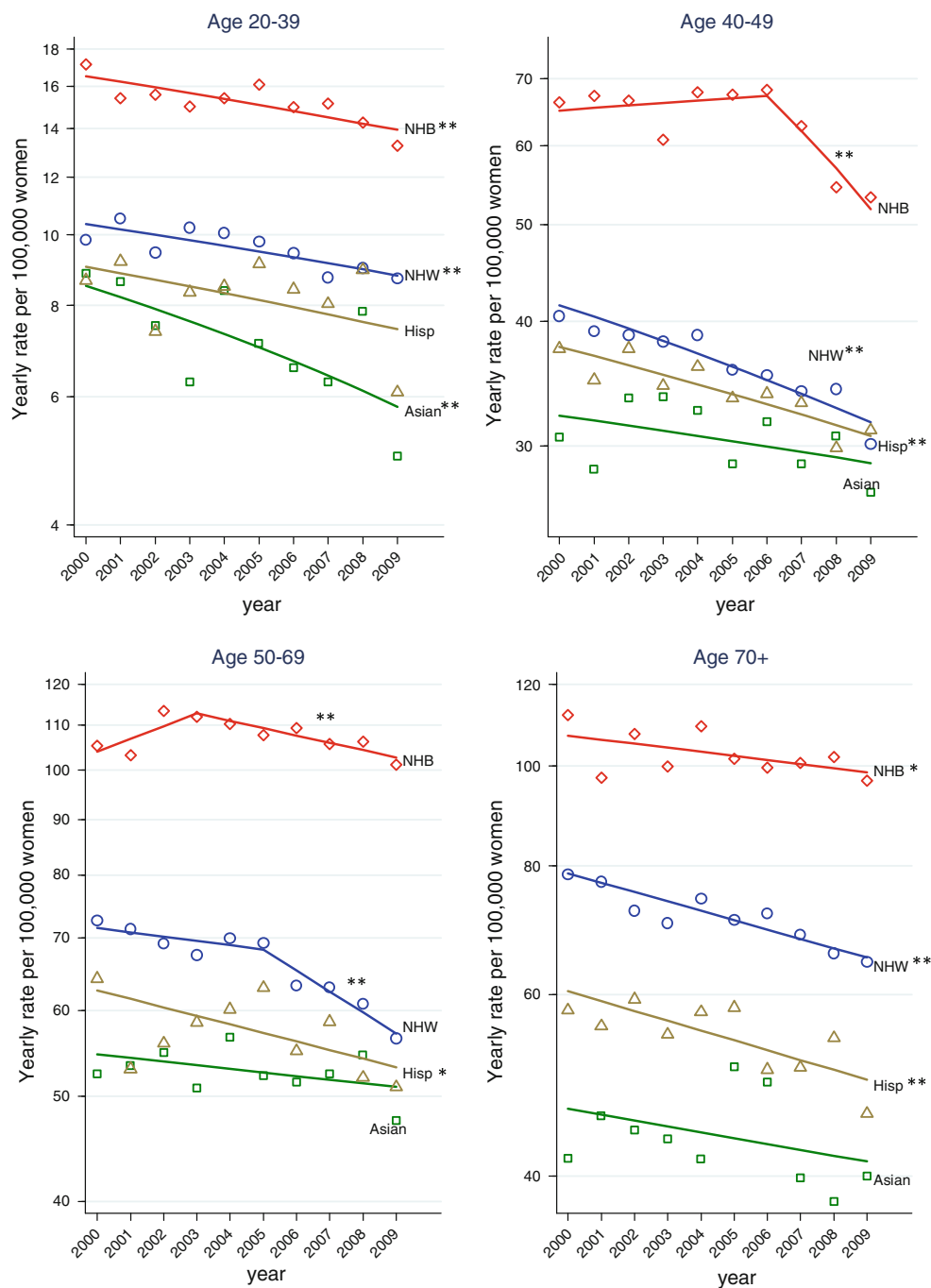
Slope with $p < 0.1$ (* $0.05 < p < 0.1$; ** $p < 0.05$):

Age 20-39: NHW 1.8 % $p = 0.001$; NHB 2.8 % $p = 0.02$; Hisp 2.0 % $p = 0.008$; **Age 40-49:** NHW 1.5 % $P < 0.0001$ %; NHB 2.4 % $p = 0.004$; Asian 2.4 % $p < 0.002$; Hisp slope II 3.0 % $p < 0.002$; **Age 50-69:** NHW slope I -4.8 % $p = 0.04$; slope II 1.4 % $p = 0.03$; NHB 2.2 % $p = 0.003$; Asian 1.2 % $p = 0.09$; Hisp slope I -0.8 % $p = 0.07$; slope II 4.3 % $p = 0.001$; **Age 70+:** NHW slope I -3.3 % $p = 0.03$; slope II 1.6 % $p = 0.006$; NHB slope II 4.5 % $p = 0.003$; Asian 1.5 % $p < 0.02$.

Fig. 4 Annual incidence rate of ER + and ER- female breast cancer in the United States with Joinpoint Regression (line of trends), by age group and race/ethnicity. Data from the U.S. National Cancer

Institute's Surveillance, Epidemiology, and End Results 18 Registries Database for cancers diagnosed between 2000 and 2009

ER- Breast Cancer



Slopes with $p < 0.1$ (* $0.05 < p < 0.1$; ** $p < 0.05$):

Age 20-39: NHW -1.8 % $p = 0.004$; NHB -1.9 % $p = 0.006$; Asian -3.7 % $p = 0.04$; **Age 40-49:** NHW -3.0 %, $p = 0.0001$; NHB slope II -8.7 % $p = 0.02$; Hisp -2.3 % $p < 0.001$; **Age 50-69:** NHW -4.5 % $p = 0.003$; NHB slope II -1.6 % $p = 0.02$; Hisp -1.8 % $p = 0.07$; **Age 70+:** NHW -2.1 % $p = 0.0001$; NHB -0.9 % $p = 0.07$; Hisp -2.2 % $p = 0.02$

NHW, non-Hispanic white; NHB, non-Hispanic black; Asian, Asian/Pacific Islander; Hisp, Hispanics

Fig. 4 continued

ended in 2005. The overall trend remained stable in Hispanics throughout the decade of 2000–2009. Previous studies reported that the breast cancer incidence stopped declining and has been relatively stable from 2005 to 2007 [7]. With the most recent 2008 and 2009 data, we observed that the incidence has started to increase in many subgroups defined by race/ethnicity, age, tumor stage and type.

For both non-Hispanic blacks and Asian/Pacific Islanders, the overall increase in breast cancer incidence appears to be driven by an increase in early-stage diseases (in situ and localized breast cancer), while the incidence of late-stage (regional and distant) breast cancer remained stable through the decade of 2000–2009. The discordant trend observed in non-Hispanic whites is possibly due to the marked reduction in the use of hormone replacement therapy in this racial group [14]. For Hispanics, although the incidence of all breast cancer appears stable between 2000 and 2009, the incidence of early-stage breast cancer significantly increased, while the incidence of late-stage breast cancer significantly decreased.

The trends for breast cancer subtype split in opposite directions, with ER-positive tumor incidence increasing while ER-negative tumor incidence decreasing in almost all age and racial/ethnic groups in recent years. This observation confirms the trend reported by earlier studies [7, 9]. Although the decreasing rate of ER-negative breast cancer is considerable and significant in all age groups, ER-breast cancer remains the highest among the blacks reported in this and previous studies [15]. On the other hand, the ER-positive cancer rate is increasing the fastest in blacks compared to all other racial groups, regardless of age. Therefore, it is likely that the molecular characteristics of breast cancer are shifting among the blacks, with fewer triple-negative subtype cancers diagnosed over time. However, comparing to other racial/ethnic groups, non-Hispanic blacks have the highest regional/distant breast cancer incidence throughout the decade, suggesting they still have the worst prognosis.

It is possible that the changes in incidence rates, especially estrogen receptor positive breast cancer, are due in part to improvements in cancer screening methods and, therefore, advance in early detection. The full-field digital mammography has replaced screen-film mammography in most breast screening programs [16]. Digital mammography has better sensitivity [17] particularly in detecting ER-positive and early-stage breast cancer such as ductal carcinoma in situ [18, 19], and was reported to be more accurate in women under the age of 50 years [20]. On the other hand, however, it is unlikely that the overall trends of incidence rates are due to changes in mammography screening rate since mammography use has not substantially changed from 2000 through 2010 [21–23], although it has increased by large magnitudes in small groups with

growing populations, such as new immigrants and Asian Americans [23]. Moreover, the U.S. Preventive Service Task Force stopped recommending routine screening for women age 40–49 in November 2009 [24, 25], which means the data we show here were not influenced by this guideline change. Increases in the incidence rate of ER+ tumor in women younger than 40 may reflect that the risk factor profile in younger generation have shifted over the years, but we could not postulate an exact reason.

This study has several strengths and limitations. We conducted a descriptive analysis of population-based cancer incidence rates from 18 SEER registries with high quality data, representing 28 % of the US population. We imputed ER status because there is significant proportion of missing data in this variable, particularly for older patients and minorities. The database was large enough for us to do subgroup analysis by multiple factors including age of diagnosis, racial/ethnicity, stage of tumor, and ER status, providing detailed information on incidence trends in small subgroups. As discussed above, part of the incidence changes may be explained by different screening technology, yet we have limited information to quantify this influence. Moreover, the incidence rates of 2001–2009 were based on populations extrapolated from the 2000 US Census, so the error in the extrapolation might bias the rate estimates. Finally, interpretation of recent breast cancer trends is limited since this study is descriptive only.

In conclusion, with the updated SEER registry data up to 2009, the recent incidence of breast cancer appears to be increasing in certain subgroups, including non-Hispanic blacks, Asian/Pacific Islanders, early-stage breast cancer, and estrogen receptor positive breast cancer. Further studies are warranted to examine possible reasons for these changes, such as changes in mammography screening methods and risk factors prevalence.

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Conflict of interest There are no potential conflicts of interest.

References

1. Jemal A, Ward E, Thun MJ (2007) Recent trends in breast cancer incidence rates by age and tumor characteristics among U.S. women. *Breast Cancer Res* 9(3):R28. doi:[10.1186/bcr1672](https://doi.org/10.1186/bcr1672)
2. Katalinic A, Rawal R (2008) Decline in breast cancer incidence after decrease in utilisation of hormone replacement therapy. *Breast Cancer Res Treat* 107(3):427–430. doi:[10.1007/s10549-007-9566-z](https://doi.org/10.1007/s10549-007-9566-z)
3. Ravdin PM, Cronin KA, Howlader N, Berg CD, Chlebowski RT, Feuer EJ, Edwards BK, Berry DA (2007) The decrease in breast-cancer incidence in 2003 in the United States. *N Engl J Med* 356(16):1670–1674. doi:[10.1056/NEJMs070105](https://doi.org/10.1056/NEJMs070105)

4. Chlebowski RT, Hendrix SL, Langer RD, Stefanick ML, Gass M, Lane D, Rodabough RJ, Gilligan MA, Cyr MG, Thomson CA, Khandekar J, Petrovitch H, McTiernan A (2003) Influence of estrogen plus progestin on breast cancer and mammography in healthy postmenopausal women: the Women's Health Initiative Randomized Trial. *JAMA* 289(24):3243–3253. doi:[10.1001/jama.289.24.3243](https://doi.org/10.1001/jama.289.24.3243)
5. Cronin KA, Ravdin PM, Edwards BK (2009) Sustained lower rates of breast cancer in the United States. *Breast Cancer Res Treat* 117(1):223–224. doi:[10.1007/s10549-008-0226-8](https://doi.org/10.1007/s10549-008-0226-8)
6. Krieger N, Chen JT, Waterman PD (2010) Decline in US breast cancer rates after the Women's Health Initiative: socioeconomic and racial/ethnic differentials. *Am J Public Health* 100(Suppl 1):S132–S139. doi:[10.2105/AJPH.2009.181628](https://doi.org/10.2105/AJPH.2009.181628)
7. DeSantis C, Howlader N, Cronin KA, Jemal A (2011) Breast cancer incidence rates in US women are no longer declining. *Cancer Epidemiol Biomarkers Prev* 20(5):733–739. doi:[10.1158/1055-9965.EPI-11-0061](https://doi.org/10.1158/1055-9965.EPI-11-0061)
8. Ehemann C, Henley SJ, Ballard-Barbash R, Jacobs EJ, Schymura MJ, Noone AM, Pan L, Anderson RN, Fulton JE, Kohler BA, Jemal A, Ward E, Plescia M, Ries LA, Edwards BK (2012) Annual Report to the Nation on the status of cancer, 1975–2008, featuring cancers associated with excess weight and lack of sufficient physical activity. *Cancer* 118(9):2338–2366. doi:[10.1002/cncr.27514](https://doi.org/10.1002/cncr.27514)
9. Anderson WF, Katki HA, Rosenberg PS (2011) Incidence of breast cancer in the United States: current and future trends. *J Natl Cancer Inst* 103(18):1397–1402. doi:[10.1093/jnci/djr257](https://doi.org/10.1093/jnci/djr257)
10. SEER-18 Surveillance, Epidemiology, and End Results (SEER) Program (www.seer.cancer.gov) SEER*Stat Database: Incidence—SEER 18 Regs Research Data, Nov 2011 Sub (1973–2009) < Katrina/Rita Population Adjustment >—Linked To County Attributes—Total U.S., 1969–2010 Counties, National Cancer Institute, DCCPS, Surveillance Research Program, Surveillance Systems Branch, released Apr 2012, based on the Nov 2011 submission
11. White IR, Royston P (2009) Imputing missing covariate values for the Cox model. *Stat Med* 28(15):1982–1998. doi:[10.1002/sim.3618](https://doi.org/10.1002/sim.3618)
12. Van Buuren S, Boshuizen HC, Knook DL (1999) Multiple imputation of missing blood pressure covariates in survival analysis. *Stat Med* 18(6):681–694
13. Kim HJ, Fay MP, Feuer EJ, Midthune DN (2000) Permutation tests for joinpoint regression with applications to cancer rates. *Stat Med* 19(3):335–351
14. Pfeiffer RM, Mitani A, Matsuno RK, Anderson WF (2008) Racial differences in breast cancer trends in the United States (2000–2004). *J Natl Cancer Inst* 100(10):751–752. doi:[10.1093/jnci/djn112](https://doi.org/10.1093/jnci/djn112)
15. Clarke CA, Keegan TH, Yang J, Press DJ, Kurian AW, Patel AH, Lacey JV Jr (2012) Age-specific incidence of breast cancer subtypes: understanding the black–white crossover. *J Natl Cancer Inst* 104(14):1094–1101. doi:[10.1093/jnci/djs264](https://doi.org/10.1093/jnci/djs264)
16. Nederend J, Duijm LE, Louwman MW, Groenewoud JH, Donkers-van Rossum AB, Voogd AC (2012) Impact of transition from analog screening mammography to digital screening mammography on screening outcome in The Netherlands: a population-based study. *Ann Oncol* 23(12):3098–3103. doi:[10.1093/annonc/mds146](https://doi.org/10.1093/annonc/mds146)
17. van Ravesteyn NT, Miglioretti DL, Stout NK, Lee SJ, Schechter CB, Buist DS, Huang H, Heijnsdijk EA, Trentham-Dietz A, Alagoz O, Near AM, Kerlikowske K, Nelson HD, Mandelblatt JS, de Koning HJ (2012) Tipping the balance of benefits and harms to favor screening mammography starting at age 40 years: a comparative modeling study of risk. *Ann Intern Med* 156(9):609–617. doi:[10.1059/0003-4819-156-9-201205010-00002](https://doi.org/10.1059/0003-4819-156-9-201205010-00002)
18. Kirsh VA, Chiarelli AM, Edwards SA, O'Malley FP, Shumak RS, Yaffe MJ, Boyd NF (2011) Tumor characteristics associated with mammographic detection of breast cancer in the Ontario breast screening program. *J Natl Cancer Inst* 103(12):942–950. doi:[10.1093/jnci/djr138](https://doi.org/10.1093/jnci/djr138)
19. Bluekens AM, Holland R, Karssemeijer N, Broeders MJ, den Heeten GJ (2012) Comparison of digital screening mammography and screen-film mammography in the early detection of clinically relevant cancers: a multicenter study. *Radiology* 265(3):707–714. doi:[10.1148/radiol.12111461](https://doi.org/10.1148/radiol.12111461)
20. Pisano ED, Gatsonis C, Hendrick E, Yaffe M, Baum JK, Acharyya S, Conant EF, Fajardo LL, Bassett L, D'Orsi C, Jong R, Rebner M (2005) Diagnostic performance of digital versus film mammography for breast-cancer screening. *N Engl J Med* 353(17):1773–1783. doi:[10.1056/NEJMoa052911](https://doi.org/10.1056/NEJMoa052911)
21. Ryerson AB, Miller JW, Ehemann CR, Leadbetter S, White MC (2008) Recent trends in US mammography use from 2000 to 2006: a population-based analysis. *Prev Med* 47(5):477–482. doi:[10.1016/j.ypmed.2008.06.010](https://doi.org/10.1016/j.ypmed.2008.06.010)
22. Miller JW, King JB, Joseph DA, Richardson LC (2012) Breast cancer screening among adult women—behavioral risk factor surveillance system, United States, 2010. *MMWR Morb Mortal Wkly Rep* 61(Suppl):46–50
23. Breen N, Gentleman JF, Schiller JS (2011) Update on mammography trends: comparisons of rates in 2000, 2005, and 2008. *Cancer* 117(10):2209–2218. doi:[10.1002/cncr.25679](https://doi.org/10.1002/cncr.25679)
24. U.S. Preventive Services Task Force (2002) Screening for breast cancer: recommendations and rationale. *Ann Intern Med* 137(5 Part 1):344–346
25. U.S. Preventive Services Task Force (2009) Screening for breast cancer: US preventive services task force recommendation statement. *Ann Intern Med* 151(10):716–726. doi:[10.1059/0003-4819-151-10-200911170-00008](https://doi.org/10.1059/0003-4819-151-10-200911170-00008) W-236